

A Comparative Analysis of PM- and MDI-QKD – Progress Report

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Abstract

Quantum Key Distribution (QKD) was first proposed in 1984 by Bennett and Brassard through their foundational Prepare-and-Measure (PM) QKD protocol, now commonly known as BB84. The literature surrounding the topic lacks a unified analysis of how BB84 compares to a more recent, security-forward protocol design approach, Measurement-Device-Independence (MDI). We seek to address this research gap by analysing metrics of (i) key rate over distance for varying topologies, (ii) cost of implementation of optimal topology, both economic and energetic, and (iii) scalability of topologies for each protocol.

1 Motivation, Aims and Research Contribution

QKD refers to any protocol in which two (or more) parties use quantum communication, together with authenticated classical communication and post-processing, to establish a shared secret cryptographic key whose security is guaranteed by laws of quantum mechanics - that is, information-theoretic security against any eavesdropper allowed by quantum physics. Since Bennett and Brassard's exposition in 1984, [1], numerous variations of QKD have been proposed, each optimised for different practical or security-related considerations.

MDI-QKD was proposed in 2012 by Lo, Curty, and Qi, [2], in the form of a novel protocol removing the need for trust in the detectors upon which protocols to that point relied. By establishing a 3rd-party relay, Charlie, to sit in between Alice and Bob, who entangles the pair's photons and performs Bell state measurements (BSMs), instead of direct photon state readings, the trust in detectors that is assumed in BB84 is removed. Given the difficulty of the problem of consistently fully classifying a device's side-channels, BB84 raises a major security vulnerability in practice. As such, [2] was proposed to nullify the detectors' security vulnerabilities in quantum communication of a secret shared key, and presents an improvement to maximum secure key communication distance.

However, in increasing security of communication, the key rate drops (e.g. [3]) in all comparisons with BB84. Initially, Lo et al. compared their MDI protocol to an entanglement-based (EB) scheme proposed by Ma et al., [4], in 2007. Since then, brief comparisons have been synthesised alongside other protocols in surveys such as [5], and refinements to the original protocol have been proposed, such as [6]. However, no unified, comprehensive comparison exists in the literature to understand how BB84 differs from the state-of-the-art of MDI-QKD with respect to: (i) key rate/distance relationships for varying topologies, (ii) economic cost of implementation of optimal topologies, and (iii) effects on cost and key rates of scaling networks (optimal topologies again) in user count.

The successful completion of this analysis may lead to a clear identification of device parameter regimes and network requirements under which our MDI-QKD protocol is preferable to BB84, and vice versa - allowing for industry and research to better identify suitable implementations of QKD for a use case. This would advance our understanding of rate and cost scaling for MDI-QKD, and bring us closer to achieving a "quantum internet", put forward in 2008, [7], by informing and simplifying selection processes for key-sharing protocols.

Less easy to quantify, and not the focus of this work, is the security comparison of the two protocols - though it is clear that security is improved significantly in the replacement of BB84 with MDI. We will refer to other works (e.g. [8], [9], [10]) in future submissions to understand these requirements.

2 Milestones and Timeline

In the Project Outline, the following milestones and delivery dates were stated to achieve the above research goals:

M1 Mathematical Modelling Complete

- 1.1 Selection and validation of a universal quantum cost function to standardise economic analysis of protocols (*end-Jan 26*),
- 1.2 Validation of modelling on top of which to build network simulations (*end-Jan 26*);

M2 Simulations Constructed

- 2.1 QuNetSim and NetSquid (first pass) simulations built and tested in Python (*end-Feb 26*);

M3 Initial Simulation Results

- 3.1 First pass at simulations of 2-node networks complete in QuNetSim for BB84 and NetSquid for MDI (*end-Mar 26*),
- 3.2 Analysis of initial results complete and drafted (*end-Mar 26*);

M4 Simulations Finalised

- 4.1 Final simulations of arbitrarily (to a limit) scaled network user counts complete (*end-Aug 26*),
- 4.2 Simulation results collated and analysed (*end-Aug 26*);

M5 Report Complete

- 5.1 All results and analysis collated into final report (*end-Aug 26*).

As things stand, **M2** is near-completion, as is **M3.1** - though noting that due to technical constraints of the library, QuNetSim has been dropped, and all early simulations will now utilise NetSquid. A re-ordering of **M3** to complete before **M1**, with all **M1-3** complete by end-Mar 26 has updated the timeline, keeping the final dates unchanged. We begin our simulations with no modelling and build it up one parameter at a time, obtaining results sequentially in each case. Presently, a simulator with zero physical modelling has been built for BB84 and its counterpart is underway for MDI, with key rate extraction and basic comparison expected to be done by 4 Mar 26.

Once this is done, physical network and device parameters such as (though not limited to) loss, dark counts, source/detector efficiency rates and basis bias will be incorporated, one at a time, comparing relative changes in cost and performance at each stage. The goal of this analysis will be to determine regimes under which comparisons of BB84 and MDI remain consistent, and to discover which parameters' modelling has the greatest effect on preferability of protocol. Cost will be analysed in similar fashion to [11], with elements of the authors' framework for modelling cost utilised, and time permitting, the novel Integer Linear Programming (ILP) optimisation methods translated for MDI. Performance will be investigated through key rate-distance graphs, relative to the fundamental PLOB bound, [12]. Various comparison scenarios will be employed, including fixing a minimum key rate and extracting an optimised cost, and fixing available budget and extracting a maximal key rate.

Once two-node networks have been simulated, efforts will be directed toward scaling up the user counts for both protocols, taking inspiration from a number of papers implementing metropolitan QKD networks (e.g. [13], [14], [15]). In the case of the multinodal network, we aim to incorporate comparisons by which we fix a topology and find marginal costs and key rates when adding a user.

Separately to the milestones, aligned with the completion of the Project Report, the upcoming Literature Review has due date 13 Mar 26. This will be adjacent to the contents of my final Report, and will seek to provide a survey of implementations of MDI-QKD, through full deployments and field trials. In particular I will focus on varieties and efficacies of varying tech platforms to host the protocol, aiming toward network implementations of greater than 2 end nodes. This will ultimately be modified and incorporated into the surrounding discussion of MDI that I will provide in my final Report. Key sources identified so far include [16] for its implementation on single atom end user nodes, [17] for its chip-based transmitters, [18] for their field deployment of an MDI network complete with BSM central node in the Netherlands and [19] for an experimental demonstration of wavelength-multiplexed MDI on integrated optical frequency combs, a future generation technology platform.

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